Remote Multi-Channel Capacitance Control System

End-to-End MATLAB Simulation — Blocks A to H

48 Channels | **BB135 Varactor** | **BPSK Comms** | **PWM Bias Gen**

Prepared for Faculty Presentation | 2025

MATLAB Simulation Report

System Overview — Signal Chain A → H

Overview

Block A

Transmitter Input

48 C targets + C-V inversion

→

Block B

ADC Encoding

10-bit + Frame assembly + CRC

→

Block C

BPSK Modulation

10 kHz carrier, 10 kbps

→

Block D

AWGN Channel

SNR = 5–25 dB noise model

Block E

Demodulation

Correlator + BER sweep

→

Block F

DAC Reconstruct

Voltage recovery from codes

→

Block G

PWM Generation

20 kHz PWM + RC LPF

→

Block H

Varactor Verify

C-V model + Error analysis

Key Parameters

ADC: 10-bit	PWM: 20 kHz	Varactor: BB135 (C_o=18pF)

Block A — Transmitter Input Stage

Block A

Varactor C-V Model

BB135 modelled as:

$$C(V) = C_0 / (1 + V/\phi)^{\gamma}$$

C_0=18pF | ϕ=0.7V | γ=0.45

Capacitance Range	5.52 pF → 16.66 pF
Bias Voltage Range	0.14 V → 8.99 V
Channels	48 (sorted ascending)
Inversion	$V_bias = \phi \cdot [(C_0/C)^{(1/\gamma)} - 1]$

Higher capacitance = lower reverse bias voltage
due to the non-linear C-V characteristic.

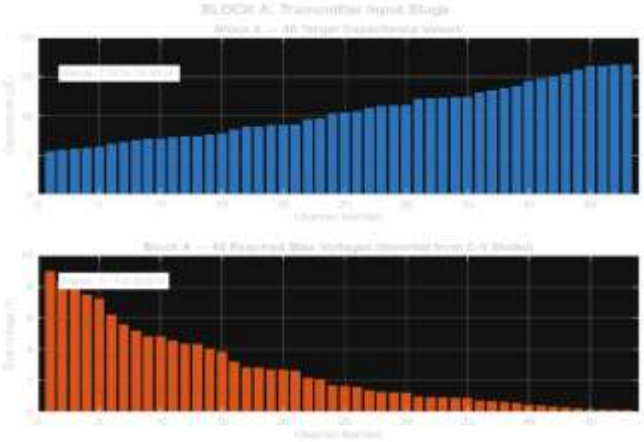


Fig 1 — Block A: 48 target capacitances (top) & required bias voltages (bottom)

Block B — ADC Encoding

Frame Structure

ADC Resolution	10 bits | dV = 9.7752 mV/LSB
Sync Word	16 bits (0xA5A5 pattern)
Data Payload	480 bits (48 ch × 10 bits)
CRC-16	16 bits (XOR checksum)
Total Frame	512 bits @ 10.0 kbps
Frame Period	51.2 ms

DATA — 480 bits (93.75%)

Quantisation error bounded within ±0.5 LSB (±4.89 mV) for all 48 channels.

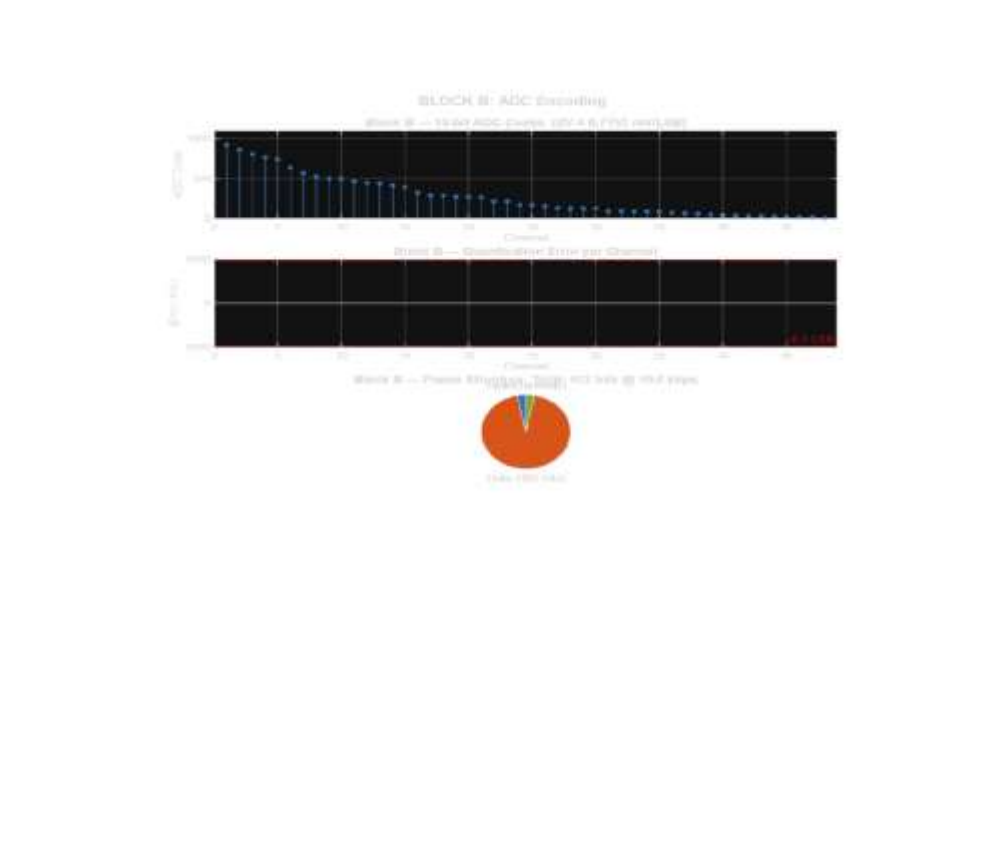


Fig 2 — Block B: ADC codes, quantisation error, and frame structure pie chart

Block C — BPSK Digital Modulation

Block C

Modulation Parameters

Scheme	BPSK (coherent)
Carrier Frequency	10.0 kHz
Symbol Rate	10.0 kbps
Samples / Symbol	50 (F_s = 500 kSa/s)
Spectral Efficiency	1 bit/s/Hz
Main Lobe BW	~20 kHz (2 × Rs)

Bit Mapping:

Bit = 1 → 0° → cos(2π·10kHz·t)

Bit = 0 → 180° → −cos(2π·10kHz·t)

PSD shows energy peaked at 10 kHz with sinc² spectral nulls at ±10 kHz.

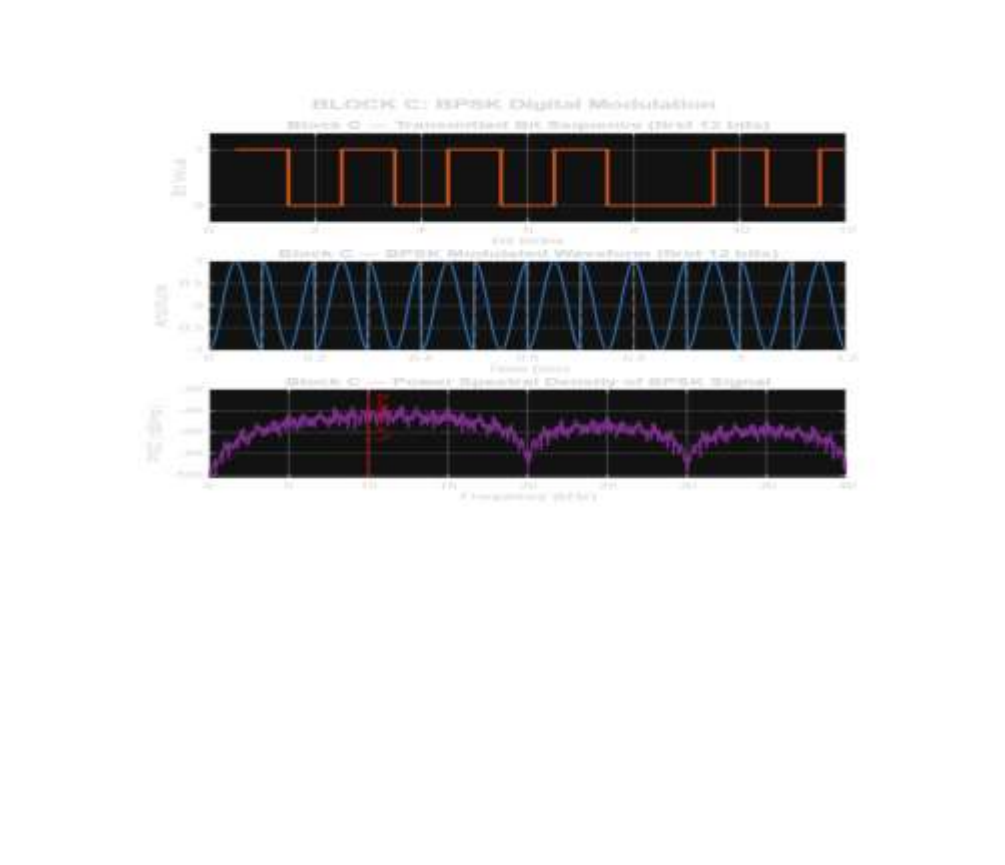


Fig 3 — Block C: Bit sequence, BPSK waveform, and power spectral density

Block D — AWGN Channel Model

Block D

Channel Parameters

Channel Model	AWGN
SNR Test Points	5, 10, 15, 20, 25 dB
Noise Distribution	Zero-mean Gaussian
Noise Variance	$\sigma^2 = E_b / (2 \cdot SNR_lin)$

Waveform Observations:

SNR = 10 dB Heavy distortion — signal barely recoverable

SNR = 15 dB Carrier structure clearly visible

SNR = 20 dB Near-clean waveform

Noise histograms confirm Gaussian distribution at each SNR level — PDF narrows as SNR increases.

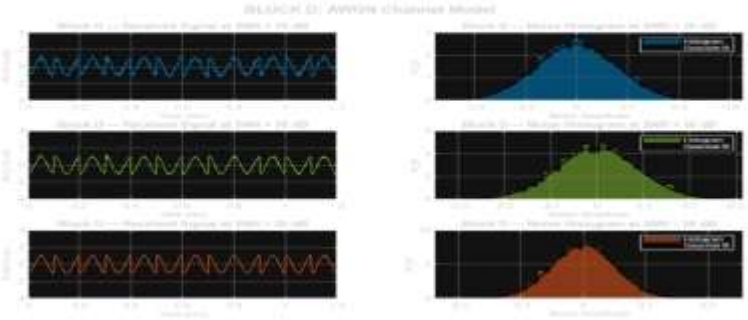


Fig 4 — Block D: Received waveforms (left) and Gaussian noise histograms (right) at SNR = 10, 15, 20 dB

Block E — Demodulation & Bit Recovery

Block E

BER Performance

SNR	BER (Theoretical)
5 dB	5.9×10^{-2}
10 dB	7.8×10^{-4}
15 dB	1.9×10^{-7}
20 dB	$\sim5\times10^{-13}$
25 dB	0 (zero errors ✓)

✓ **CRC-16 PASSED — All 512 bits correctly recovered at 25 dB SNR**

Demodulator: coherent correlator matched to $\cos(2\pi\cdot10\text{kHz}\cdot t)$.
Decision threshold = 0.

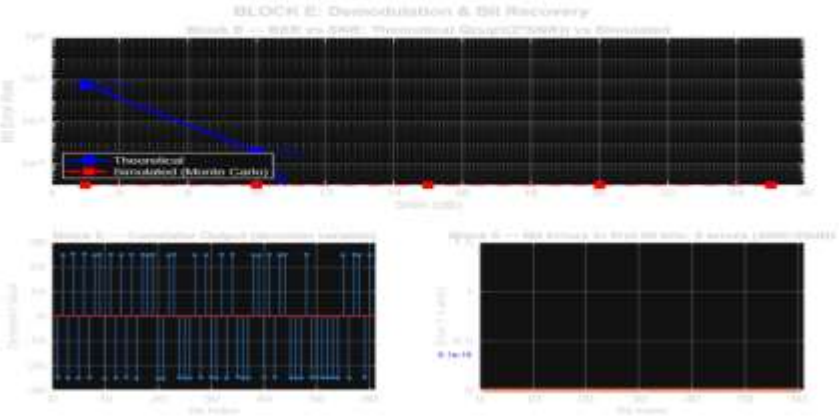


Fig 5 — Block E: BER vs SNR curve, correlator output, and bit error indicator at 25 dB

Block F — DAC Reconstruction

Reconstruction Performance

Max Voltage Error	4.870 mV < 0.5 LSB ✓ PASS
RMS Voltage Error	2.529 mV
Error Source	ADC quantisation only
Max DNL	2.0000 LSB
Max INL	512.0000 LSB (expected — uniform Q)

All 48 channels within ±0.5 LSB bounds
→ Near-perfect voltage reconstruction

INL accumulates linearly — inherent behaviour of a uniform quantiser, not an additional non-linearity.

Zero link errors at 25 dB SNR → reconstructed codes = transmitted codes exactly.

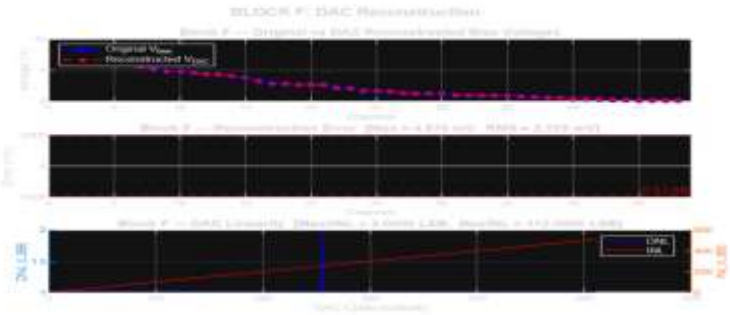


Fig 6 — Block F: V_orig vs V_recon overlay, reconstruction error, and DAC DNL/INL

Block G — PWM-Based DC Bias Generation

Block G

RC Filter Design

PWM Frequency	20 kHz
RC Cutoff (f_c)	1 kHz (5% of PWM freq)
RC Time Constant	0.1592 ms | R = 10 kΩ
Attenuation @ 20kHz	−26 dB (single-pole)

Ripple Results:

Ch 6 (D=62.3%) → 1539.9 mV pp

Ch 24 (D=17.1%) → 2181.6 mV pp

Ch 42 (D=3.5%) — WORST → 2278.6 mV pp

✗ FAIL — Max 2278.6 mV pp vs 50 mV spec
→ RC filter insufficient — root cause of C error

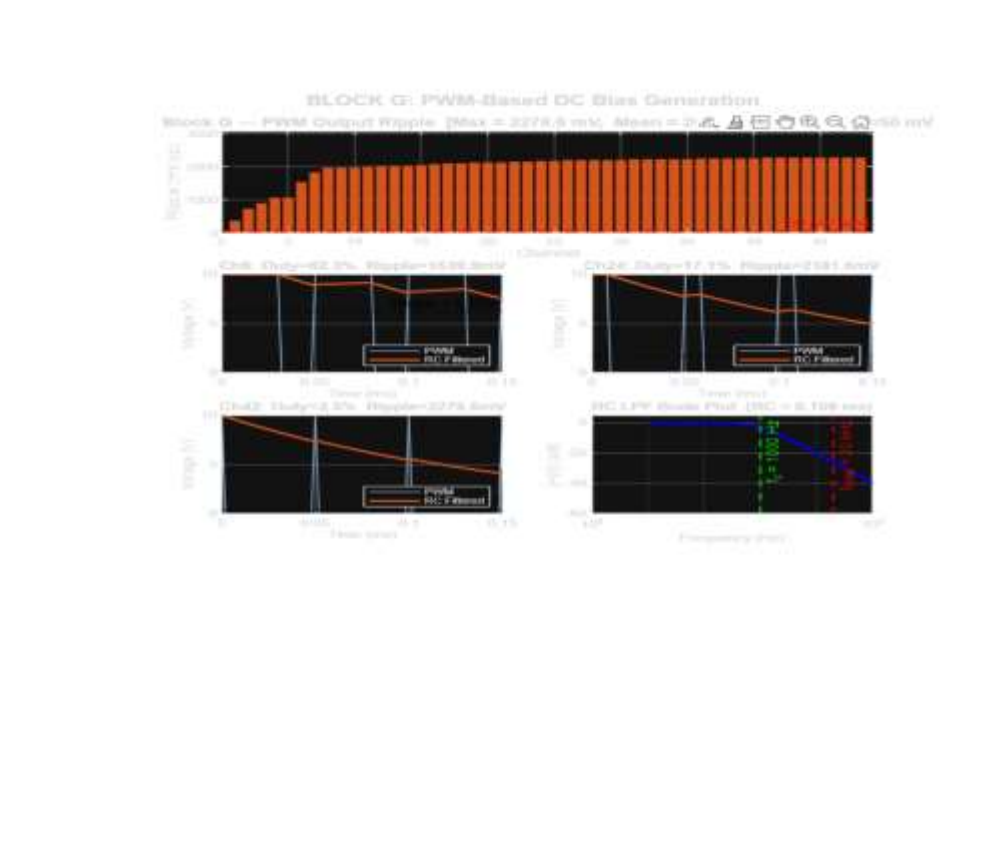


Fig 7 — Block G: Ripple per channel, PWM waveforms Ch6/24/42, and RC Bode plot

Block H — Varactor Characterisation & Verification

Block H

End-to-End Error Analysis

Max Capacitance Error	9.690 pF
RMS Capacitance Error	5.090 pF
Max Error (%)	58.38% (Channel 48)
Mean Error (%)	34.28% (all 48 channels)
Best Channel Error	<3% (Channels 1–3)

Root Cause:

→ PWM ripple → incorrect mean voltage at RC output

→ Voltage error → amplified by non-linear C-V curve

→ Low duty-cycle channels (<10%) worst affected

Digital chain (A–E) correct ✓ | Analog chain (G) requires redesign ✗

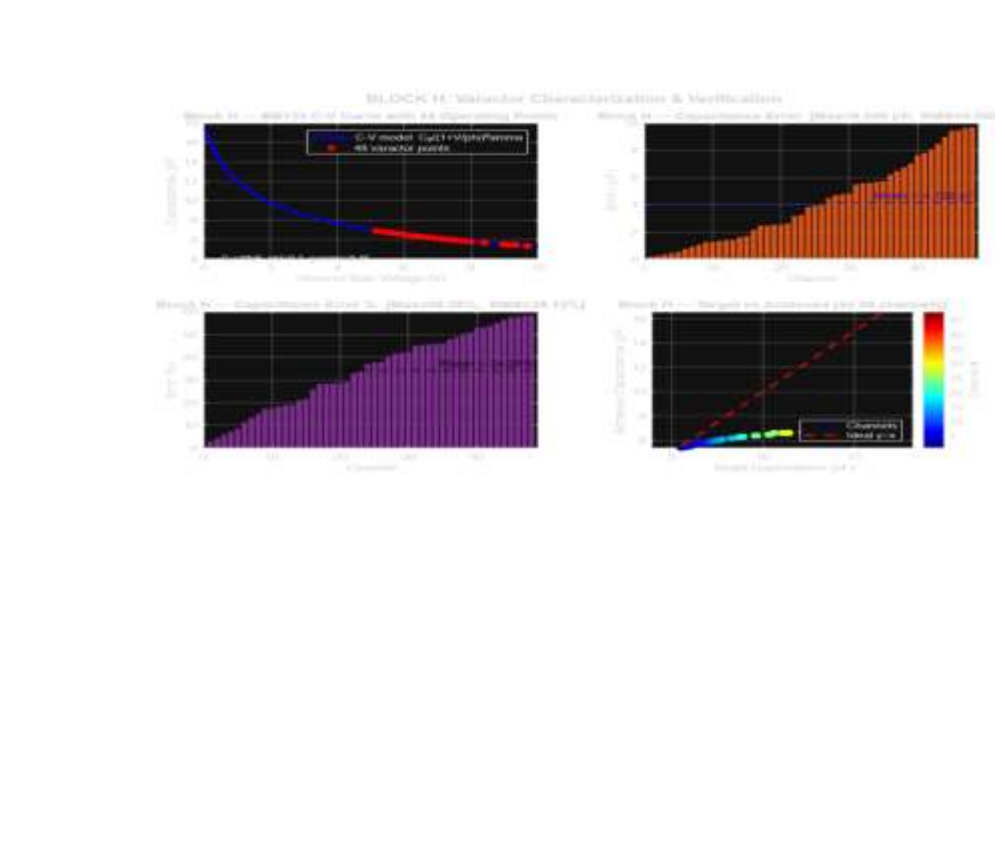
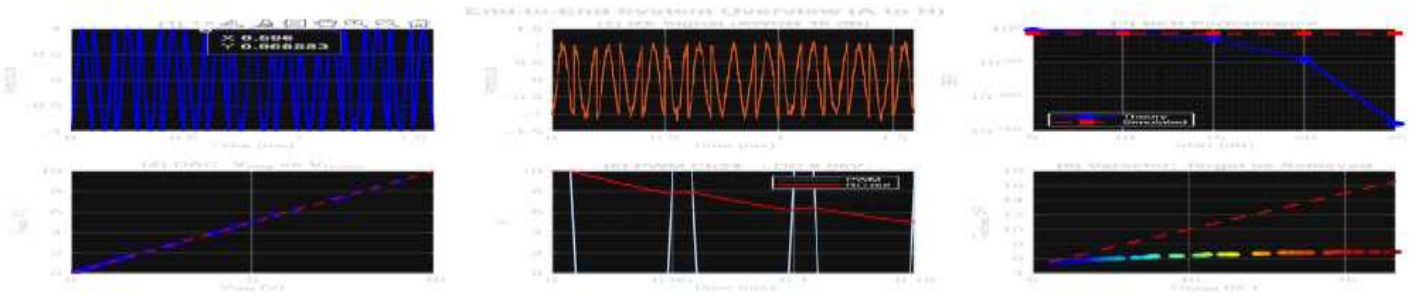


Fig 8 — Block H: C-V curve operating points, absolute error, % error, target vs achieved scatter

End-to-End System Overview — Blocks A to H

Overview

Complete A → H Signal Chain Validated



(1) TX BPSK · (2) RX @15dB · (3) BER · (4) DAC V_orig vs V_recon · (5) PWM Ch24 RC output · (6) Varactor target vs achieved

Full 48-Channel End-to-End Results — Channels 1 to 29

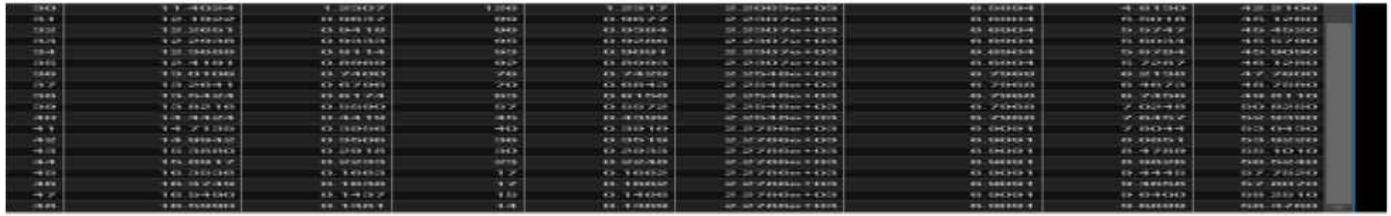
Fig 10

FIGURE 10 — Full 48-Channel End-to-End Results Table

Ch	C target (pF)	V bias (V)	ADC Co...	V recon (V)	Ripple (mV...	C achieved (pF)	Error (pF)	Error (%)
1	5.5400	8.9010	1000	8.8892	974.9100	5.5540	0.1340	2.4200
2	6.6270	8.8011	850	8.8071	709.8500	6.6450	0.0180	0.2800
3	6.6802	7.9100	800	7.8085	900.0800	6.7200	0.0398	0.5900
4	6.8645	7.4760	760	7.4760	1.082e+03	6.8260	0.0279	0.4070
5	8.0360	7.2080	740	7.2080	1.062e+03	8.0260	0.0400	0.4970
6	8.4407	6.8274	657	6.8040	1.050e+03	8.8800	0.3740	4.4410
7	8.7011	6.5080	572	6.5014	1.050e+03	8.8017	0.0804	0.9170
8	8.8000	6.1002	508	6.1000	1.024e+03	8.8800	1.0800	10.0000
9	7.8020	4.8080	480	4.8087	1.024e+03	8.8040	1.0020	12.8710
10	7.0070	4.8061	400	4.8087	1.024e+03	8.8040	1.8030	12.8740
11	7.3071	4.6050	460	4.6060	1.080e+03	8.0801	1.0801	10.0710
12	7.5001	4.5470	445	4.5400	2.000e+03	8.0800	1.0800	10.6700
13	7.4170	4.3180	432	4.3080	2.000e+03	8.1800	1.4800	19.0700
14	7.8070	4.0070	414	4.0000	2.000e+03	8.0800	1.0800	14.4000
15	7.7000	3.8860	384	3.8614	2.060e+03	8.0704	1.0700	13.8610
16	8.0070	3.7100	360	3.7100	2.070e+03	8.1017	1.1017	20.0000
17	8.6700	3.3400	301	3.3440	2.100e+03	8.2314	2.4440	28.1700
18	8.8000	3.0007	300	3.0040	2.100e+03	8.1014	2.4060	30.3000
19	8.8001	2.7080	272	2.7072	2.110e+03	8.0147	2.5140	29.4700
20	8.8000	2.7000	272	2.7072	2.110e+03	8.0147	2.5147	29.0000
21	8.8100	2.6080	270	2.6080	2.110e+03	8.0147	2.5040	29.1700
22	8.0000	2.1007	221	2.1005	2.100e+03	8.4010	2.1014	32.0070
23	8.4800	2.0080	210	2.1017	2.100e+03	8.4010	2.2400	33.0000
24	10.5800	1.7100	170	1.7107	2.100e+03	8.4014	2.0700	32.1700
25	10.4147	1.4010	140	1.4010	2.100e+03	8.4014	2.0010	32.0000
26	10.0000	1.1200	101	1.1200	2.080e+03	8.0004	4.0010	32.0000
27	11.0071	1.0070	140	1.0080	2.060e+03	8.0004	4.4070	40.4000
28	11.2700	1.2777	151	1.2000	2.040e+03	8.0004	4.0003	41.1010
29	11.2470	1.2010	168	1.2010	2.040e+03	8.0004	4.2400	41.1010

Full 48-Channel End-to-End Results — Channels 30 to 48

Fig 11



Channel 48 represents the worst case:

C_target = 16.60 pF	V_bias = 0.138 V	ADC code = 14
C_achieved = 6.91 pF	Error = 9.69 pF (58.38%)	

System Performance Summary

METRIC	RESULT	STATUS
ADC Resolution	10 bits | 9.7752 mV/LSB	—
Frame Size	512 bits @ 10 kbps	—
Modulation	BPSK (coherent)	—
BER @ 25 dB (theoretical)	$\sim 3\times10^{-20}$	PASS ✓
BER @ 25 dB (simulated)	0.00 — zero errors	PASS ✓
CRC-16 Result	PASSED	PASS ✓
Max V Recon Error	4.870 mV < 0.5 LSB	PASS ✓
RMS V Recon Error	2.529 mV	PASS ✓
Max PWM Ripple	2278.6 mV pp	FAIL ✗
Max C Error	9.69 pF (58.38%)	FAIL ✗

Conclusions & Recommendations

Conclusion

Conclusions

Digital Chain (A–E)

Zero bit errors @ 25 dB SNR | CRC PASSED | DAC within ±0.5 LSB

Analog Chain (G)

RC filter gives only −26 dB at 20 kHz | 2.28 V pp ripple | Spec FAILED

Root Cause

PWM ripple → voltage offset → non-linear C-V amplifies error up to 58%

Recommendations

1. Lower RC cutoff to ≤100 Hz (C = 1.6 µF, R = 1 kΩ) to achieve ≤50 mV ripple
2. Use 2nd-order Sallen-Key active LPF for −40 dB/decade roll-off
3. Increase PWM to 200 kHz — reduces filter order needed for same attenuation
4. Add closed-loop feedback via capacitance-to-digital converter (e.g., AD7747)
5. For duty cycle <5% channels: use dedicated voltage DAC (e.g., DAC8568)
6. Increase ADC to 12 bits (dV = 2.4 mV/LSB) for finer voltage precision

Thank You

Remote Multi-Channel Capacitance Control System

48-Channel | BPSK 10 kbps | 10-bit ADC | BB135 Varactor

Digital chain: PASS ✓ | PWM analog chain: FAIL ✗ (filter redesign needed)

Faculty Presentation | 2025